POWER LED

I = 3 mA

Mounting Holes

Decoupling capacitor

Analog voltage Filtering

Reset circuit

Crystal circuit

the crystal load capacitance is 8pF, and assuming Cstray=4pF, then:
CX1 = CX2 = 2(9pF - 5pF) = 8pF
ECS-160-9-33B-CTP-TR

Boot pin

USB connect

Microcontroller

STM32F103CBTx

J-Link Connector

Voltage regulation

LDL1117S33R

Debug led

ESD Protection circuit

EXTERNAL PIN HEADERS

CURRENT MEASUREMENT

INA2181A1IDGSR

Gain = 20

Supply 6 Volt max

V_Out = (I_Load x R_Sense x GAIN) + V_Ref
V_Out = (I_Load x R_Sense x 20) + V_Ref
V_Out = (4 x 0.020 x 20) + 1.65 = 3.25 volts maximum

INA2181A3QDGSRQ1

Gain = 100

Supply 6 Volt max

V_Out = (I_Load x R_Sense x GAIN) + V_Ref
V_Out = (I_Load x R_Sense x 100) + V_Ref
V_Out = (1A x 0.032 x 100) = 3.1 volts maximum

RELAY Circuit-4A

SRD-05VDC-SL-C

Vin * 0.25 <= 3.3 V
12 Volt max
MMBT3904-ND BJT transistor:
Ib(Sat) = 1mA , IC(Sat) = 10mA
Ib = (3.3 - 0.7) / 1k = 2.6mA
Ic1 = (5v- 2v - 0.2v) / 1K = 3 mA
Ic2 = 87 mA from coil circuit

70uA current for voltage sense

Vin * (47/124+47) <= 3.3 V
12 Volt max

12/(124K+47K) = 70uA current for voltage sense

ADC Protection

F_CutOff
= 1/(2(PI)(R)(C))
= 1/(2*3.14*(150+10)*10n)
=- 99.5KHz

External VDD subbly

External Analog Input

External thermistor sensor

USB connect-1A maximum

USB connect indicator

CAN transciever

TJA1051T-3

5.5 volts MAX

MAX Current is 193mA + MCU current

QUANTTRONIX LLC

Sheet: /
File: usb_relay_control.kicad_sch

Title: USB relay circuit

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